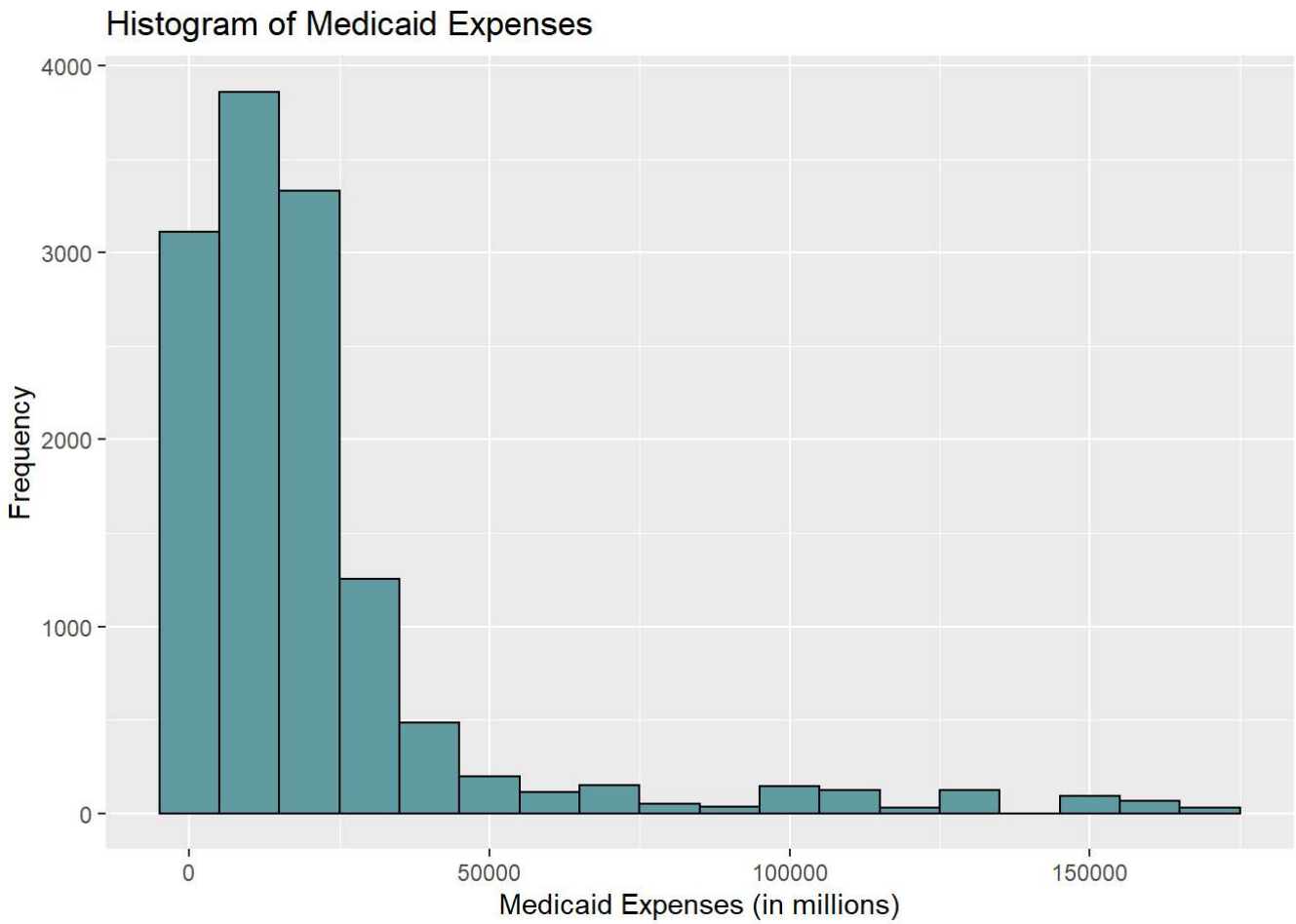
Chronic_Kidney_Analysis

2025-07-03

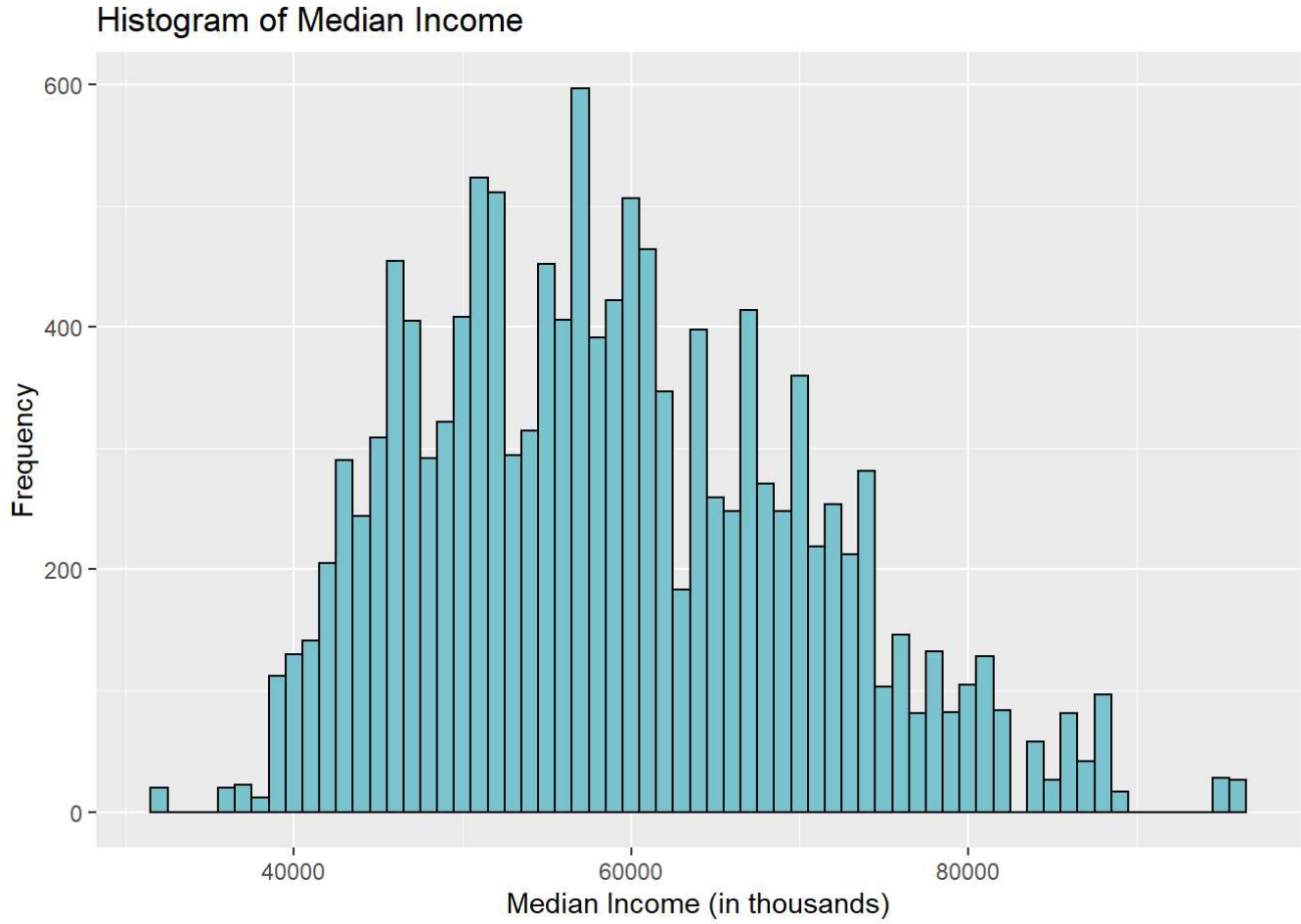
Medicaid Expenses Distribution

```
ggplot(final_df, aes(x = Medicaid_Expenses)) +
  geom_histogram(binwidth = 10000, fill = "cadetblue", color = "black") +
  labs(title = "Histogram of Medicaid Expenses", x = "Medicaid Expenses (in millions)", y = "Fre
quency")
```



Median Income Distribution

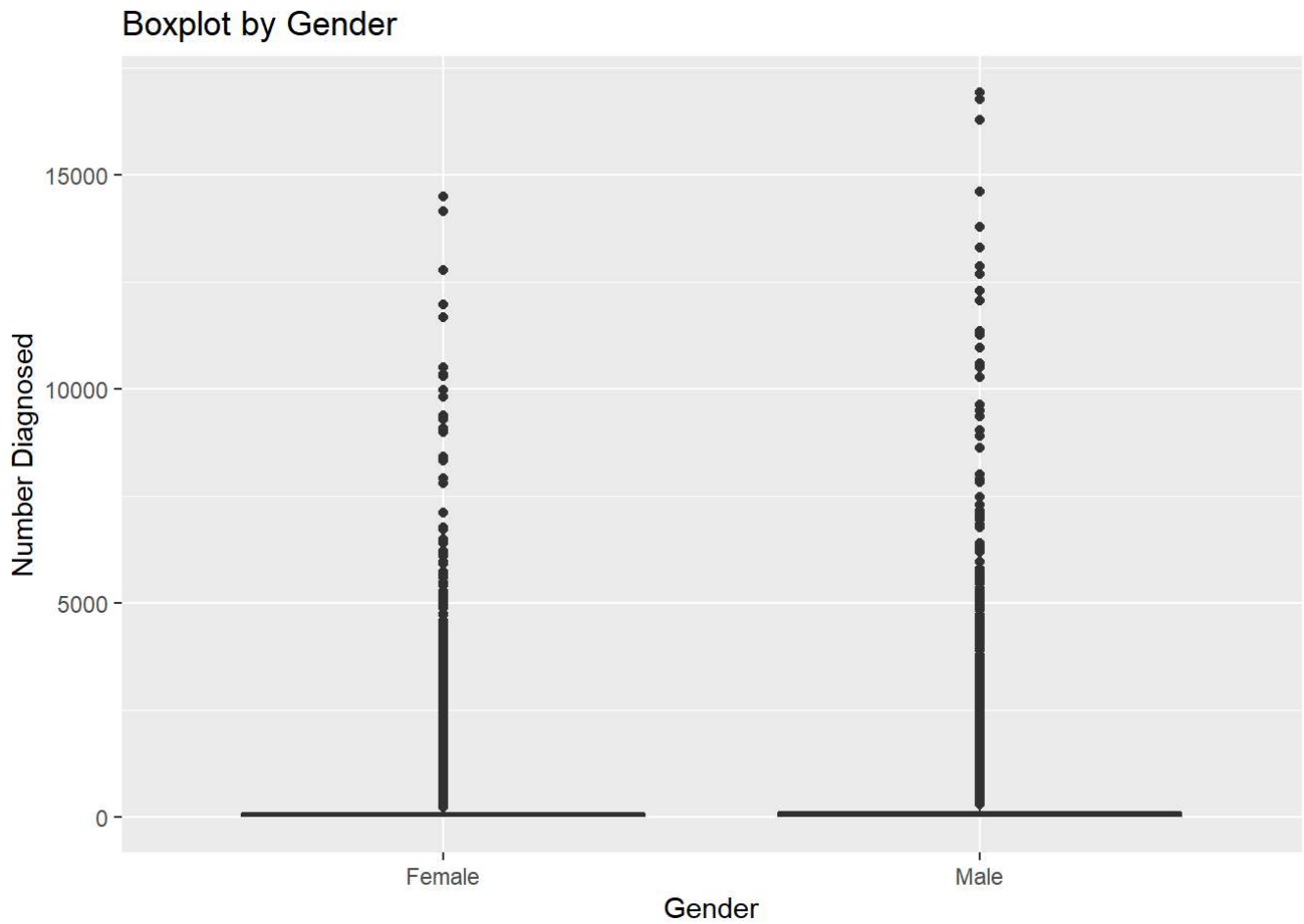
```
ggplot(final_df, aes(x = Median_Income)) +
  geom_histogram(binwidth = 1000, fill = "cadetblue3", color = "black") +
  labs(title = "Histogram of Median Income", x = "Median Income (in thousands)", y = "Frequenc
y")
```



Diagnosed Cases by Gender

```
# filter to gender
gender <- final_df %>% filter(StratificationCategory1 == "Gender")

# number diagnosed by gender
ggplot(gender, aes(x = Stratification1, y = Number_Diagnosed)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot by Gender", x = "Gender", y = "Number Diagnosed")
```



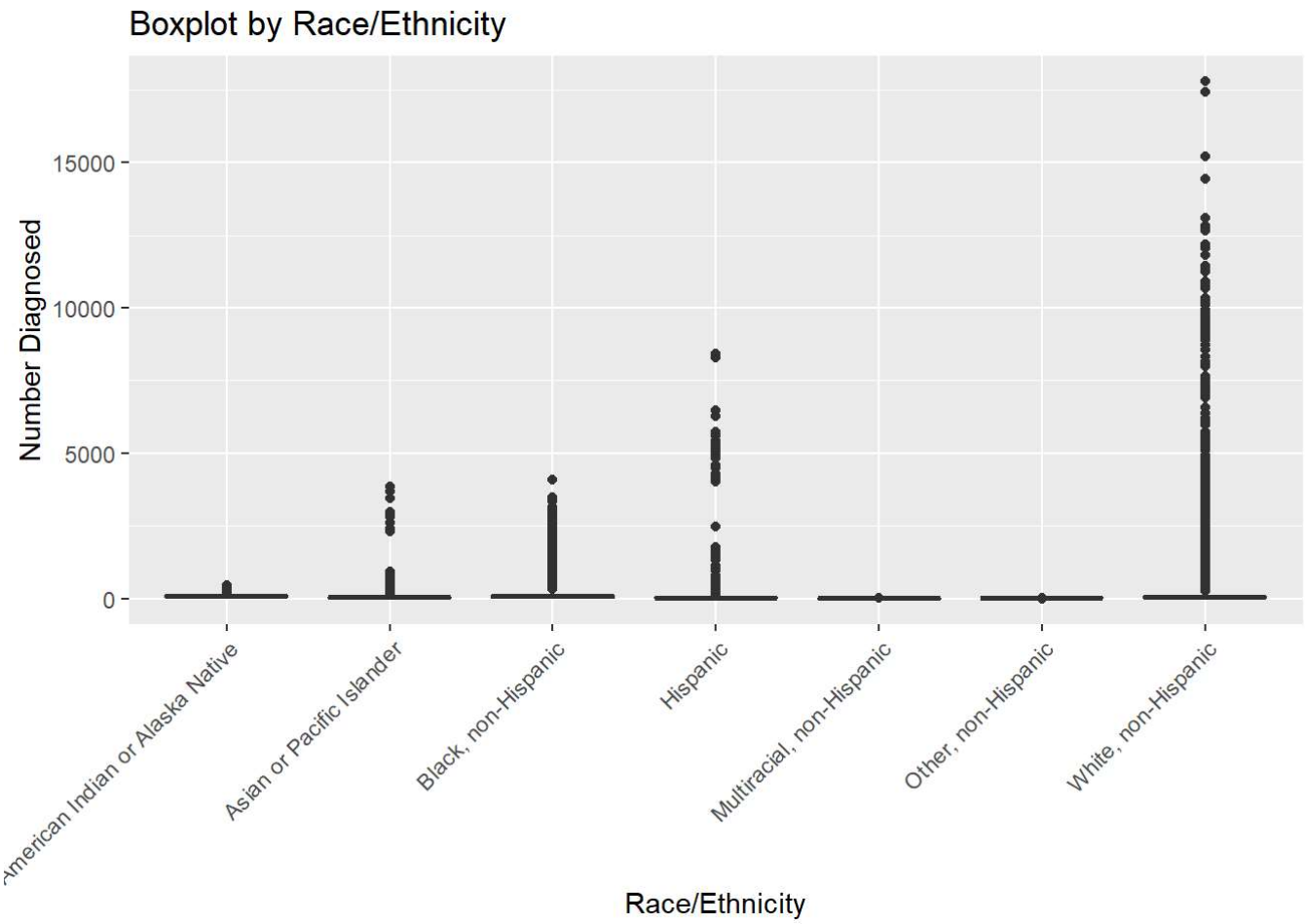
Insights

1. Most values are very low: The bulk of the data is concentrated near zero for both genders. This is evident because the box (interquartile range) is compressed at the bottom, indicating that most counties/zip codes/areas have relatively low diagnosis counts.
2. Presence of extreme outliers: Both Female and Male groups have many extreme outliers — points far above the box and whiskers. These represent areas with unusually high numbers of diagnoses, possibly densely populated or high-incidence regions.
3. No major difference between genders: The shape, median, and spread of the boxplots for Female and Male appear very similar. This suggests there may be no strong gender-related disparity in the number of diagnoses based on this boxplot alone.

Diagnosed Cases by Race/Ethnicity

```
#filter to race/ethnicity
race <- final_df %>% filter(StratificationCategory1 == 'Race/Ethnicity')

# number diagnosed by race/ethnicity
ggplot(race, aes(x = Stratification1, y = Number_Diagnosed)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot by Race/Ethnicity", x = "Race/Ethnicity", y = "Number Diagnosed")+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

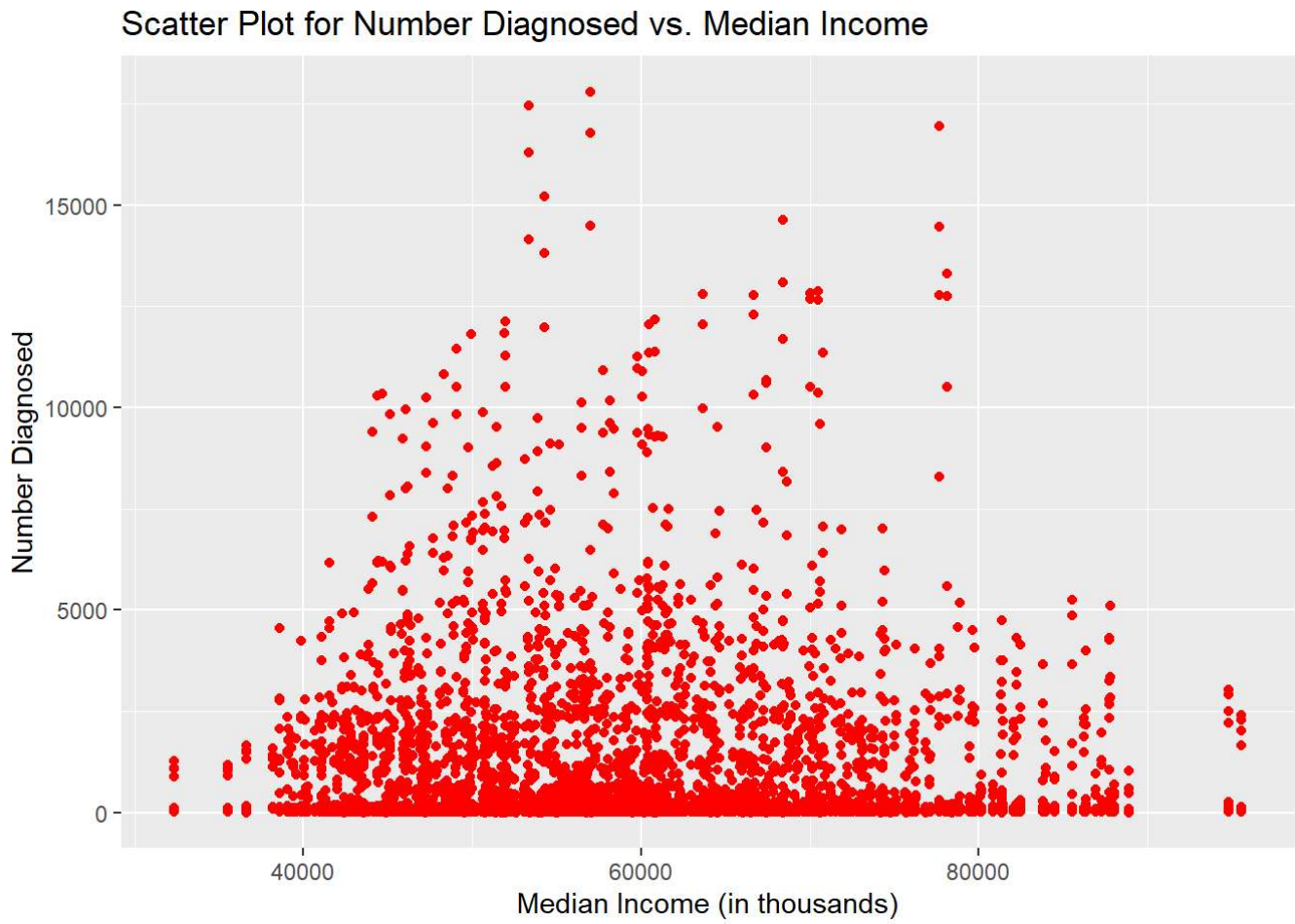


Insights

1. Most groups have low diagnosis counts overall: Like the gender plot, the majority of values across all race/ethnicity groups are concentrated near zero. This is indicated by the compressed box and whiskers at the bottom of each category.
2. White, non-Hispanic group shows the highest spread and outliers: This group has the widest range of diagnosis counts and the most extreme outliers (up to 400,000+), suggesting: Larger population representation Possibly more regions or counties with high case counts Or greater variation within the group
3. Hispanic and Black, non-Hispanic groups also show relatively high diagnosis counts: These two groups have longer whiskers and more upper outliers compared to others, though less than the White, non-Hispanic group. This could reflect higher burden or concentration in certain areas.
4. Other racial/ethnic groups show more tightly clustered counts: Groups like Asian or Pacific Islander, American Indian or Alaska Native, Multiracial non-Hispanic, and Other non-Hispanic have relatively lower and tighter diagnosis counts. This might reflect smaller populations, underreporting, or lower incidence.

Diagnosed number vs median income

```
# scatterplot showing number diagnosed against median income
ggplot(final_df, aes(x = Median_Income, y = Number_Diagnosed)) +
  geom_point(color = "red") +
  labs(title = "Scatter Plot for Number Diagnosed vs. Median Income", x = "Median Income (in tho
usands)", y = "Number Diagnosed")
```

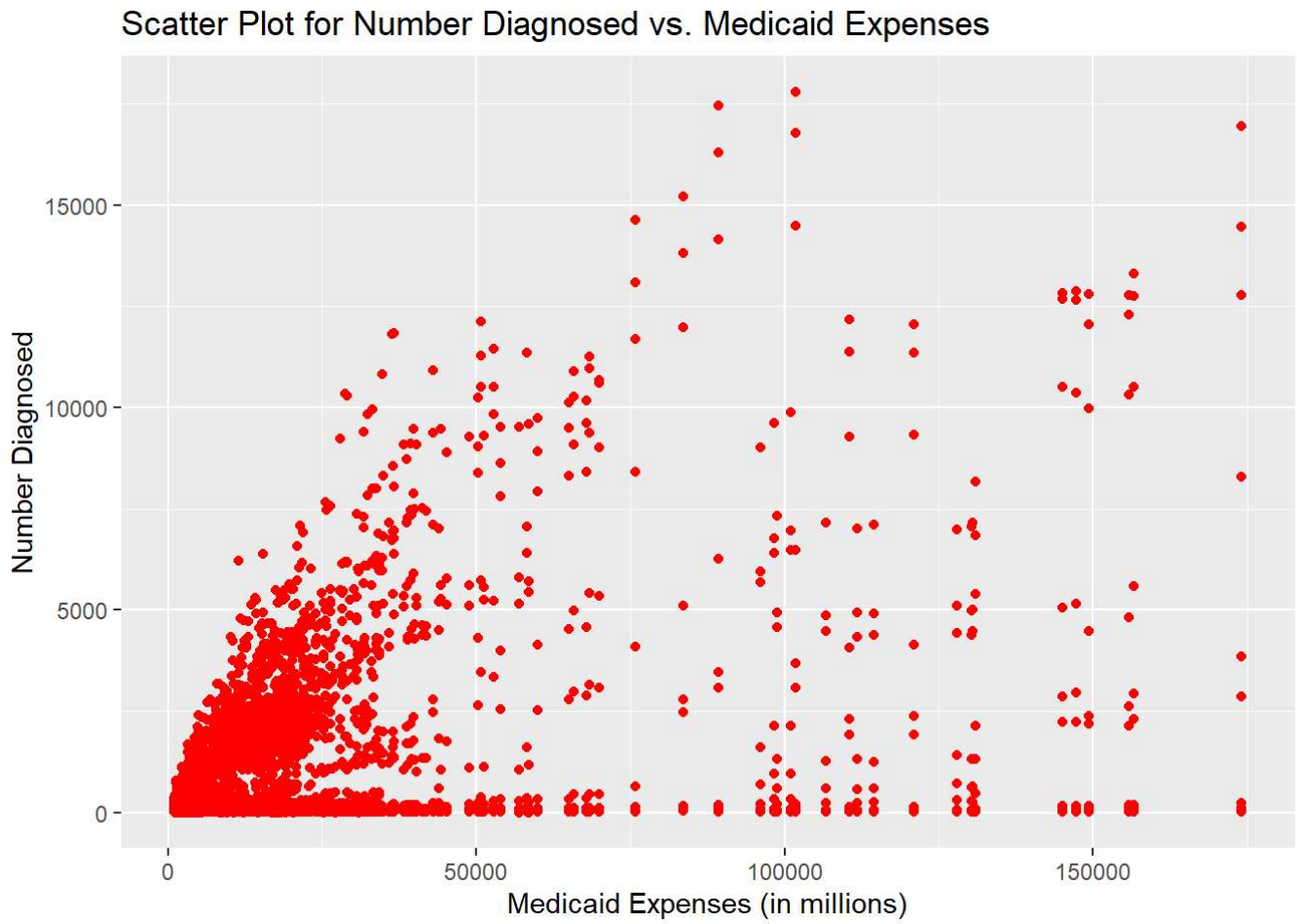


###Insights###

1. Negative correlation trend: There is a visible downward trend: as median income increases, the number of diagnosed cases tends to decrease. This suggests that lower-income areas may be experiencing higher diagnosis counts.
2. High variability at low-income levels: At income levels around $30,000–$60,000, there is wide spread in the number diagnosed (ranging from near 0 to over 300,000). This indicates that while low-income areas tend to have high counts, some still have low counts — suggesting other influencing factors.
3. Upper-income areas show fewer diagnoses: For median incomes above $80,000, diagnosis counts are consistently low, with fewer and smaller outliers. This supports the idea that wealthier communities may have fewer diagnoses, possibly due to better access to care, preventive measures, or lower disease prevalence.

Diagnosed number vs medicaid expense

```
# scatterplot showing number diagnosed against medicaid expenses
ggplot(final_df, aes(x = Medicaid_Expenses, y = Number_Diagnosed)) +
  geom_point(color = "red") +
  labs(title = "Scatter Plot for Number Diagnosed vs. Medicaid Expenses", x = "Medicaid Expenses
(in millions)", y = "Number Diagnosed")
```



###Insights###

1. Curved cluster of points on the left (0–50,000): Most data points lie in the lower Medicaid expenses range (up to 50,000), and within this range, the number diagnosed increases with spending. This suggests a positive relationship between Medicaid spending and diagnosis count in lower-spending areas, possibly because: Areas with more diagnosed cases receive more funding, or Spending helps uncover and treat more diagnoses.
2. Flat cluster at high Medicaid spending levels: Beyond $75,000+ in expenses, the number diagnosed becomes flat or scattered and lower overall. Some high-spending areas show surprisingly low diagnosis counts, which could indicate:
3. Non-linear relationship: The plot does not show a consistent linear pattern across the full range. There's a rising trend early on, but it plateaus or declines in higher ranges — suggesting diminishing returns or varying context for high Medicaid expenses.

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correlation analysis

cor_matrix <- final_df %>% select(where(is.numeric)) %>% cor(use = "complete.obs")

corrplot::corrplot(cor_matrix, method = "circle")